FROM ABSTRACTS TO INSIGHTS:

Clustering Scientific papers

Task 3: Categorizing Trends in Science

IU International University of Applied Sciences

Case Study

Degree Programme: Data Science

Machine Learning – Unsupervised Learning and Feature Engineering DLBDSMLUSL01

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List of Abbreviations

AI	Artificial Intelligence
NLP	Natural Language Processing
TF-IDF	Term Frequency-Inverse Document Frequency
SVD	Singular Value Decomposition
PCA	Principal Component Analysis
MDS	Multidimensional Scaling
GMM	Gaussian Mixture Models
LLMs	Large Language Models
EDA	Exploratory Data Analysis

1. Introduction

In this case study we will look at how the company can find research fields that they may want to collaborate with. By doing this they need to know what fields of research are currently being looked into.

As there are too many research papers to go through and look at what people are researching. Usually all the key information can be found within a papers abstract. Which leaves us with a huge amount of data that we would have to process with machine learning to get any useful information out of.

So I will be looking at an archive of Arxiv papers. (Data set that can be found on kaggle that was released by Cornell university). I can then use this to create an insight into what science is currently researching. The full data set is about 1.7 million papers long which is to much data to work with and would take too long. So I will be working with 150,000 papers.

I decided to only look at papers that were released from 2023-2025 as well. Because we want to see what science is currently researching not what was researched years ago. We want to see what topics are being researched that the company could collaborate with in the future.

By creating these clusters of papers that are similar we can start to get some insights on what science is doing. What topics are they looking into that could benefit our business. Some questions that I want to answer are what fields of science are growing.

What fields are already popular.

What topics stand out that may be beneficial for our company to collaborate with.

Using unsupervised machine learning to help find patterns in the data and cluster similar papers together. The result will give us some insight that we can use for our company.

2. Methodology: From Raw Data to Insights

To approach this problem, I believed that following an iterative procedure of looking through the data exploration, preprocessing, build features engineering and apply clustering would allow me to find an answer. By building out a way to convert my raw data into something that can be clustered I was able to get to my result.

It's also helpful to know how to build out our pipeline properly from the start. Throughout this problem I found myself going back through many parts of my pipeline multiple times. So being able to have my pipeline built out in a manner where I can understand what changes were made was beneficial.

2.1 Data Source and Sampling

The arXiv database has approximately 1.7 million papers. For computational reasons and space, it's not feasible to use the entire dataset. When trying to vectorize text and run clustering models.

What would we consider current scientific trends? After a bit of research I believe anywhere from 2-5 years would be a good amount. For simplicity and to make the data easy to understand I chose to look at the past 3 complete years.

I filtered my dataset to only include papers from the years 2023-2025. This reduced my dataset by quite a bit but still had plenty of useful information for our purposes. It was still too large of a dataset.

I took a random sample of 150,000 papers from this new dataset. I used fix set of seed to have reproducible results. 150,000 seemed like a good number because I could still get a good variety of topics and wouldn't take too long to process through my machine learning model.

The JSON lines format allowed me to not have to load everything into memory.

2.2 Exploratory Data Analysis and Data Quality

Before I began using any models, I wanted to see what my data looked like. As mentioned before this is a very important step to know what your data contains and can help you make decisions on what to do in following steps. Many decisions were made from my Exploratory Data Analysis (EDA) when preprocessing and modelling.

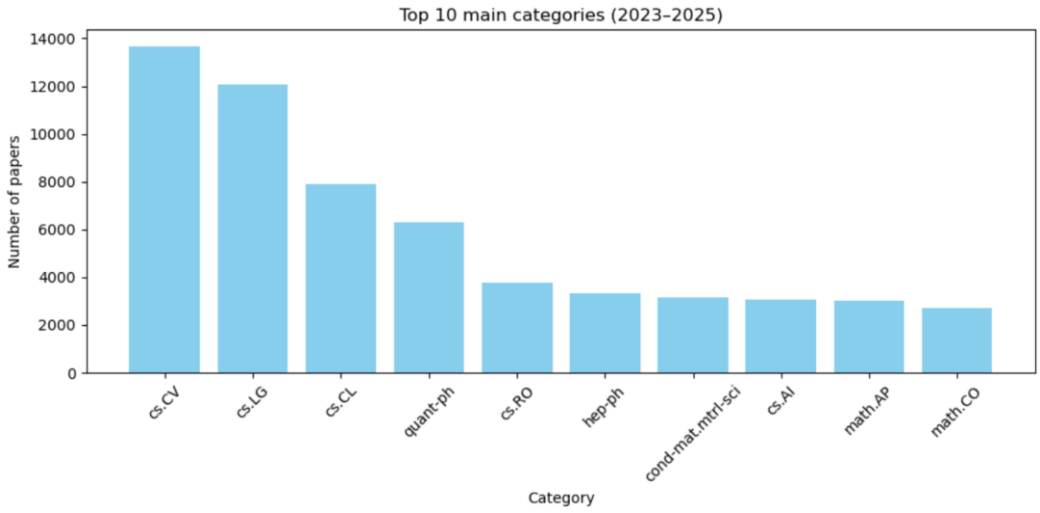


Figure 1: Top 10 main categories (2023-2025) before modelling

I analysed how many papers there are each year and what categories are the most common. As well as abstract length and any missing/empty values. There were 39,955 papers published in 2023, 47,469 in 2024 and 62,576 in 2025. Which shows that more papers are being published each year. There were no missing or empty values for abstracts and only one abstract with just 3 characters long which was removed.

I decided not to balance the number of categories as I want the data to be as realistic as possible.

From my analysis we can see that there are many categories that relate to AI. These include Machine Learning and Natural Language Processing which are already high amounts within our data set.

2.3 Text Preprocessing and Feature Engineering

Because of our data consists of abstracts that are in text format we need to represent them in numbers.

We can do some pre-processing on the text by converting it all to lower case. Removing special characters with regex and finally removing English stop words and words that occur often in the dataset such as equation, theorem, method etc. We want to remove these because they take up space in our data but don't give us any insight to the topic of the paper.

I looked at different ways to represent text as vectors before deciding on TF-IDF. One option was Bag - of - Words but I didn't think this would work as good because it doesn't look at how much each word matters per document. N-grams was another option but since our dataset was already going to be big with only words I decided against this. So, I decided to use Term Frequency-Inverse Document Frequency or TF-IDF for representing the data.

With TF-IDF we can see what terms are more important in each document but not to prevalent throughout all the text. This works well for finding topics in text data. I decided to use the top 8000 terms in my vocabulary and dropped the low frequency and very high frequency terms. So now we have each abstract as a vector in 8000 dimensions. Where each dimension is a term in the vocabulary and how important that word is in the document.

Now that we have our data represented in numbers, we can compare papers and see if they are similar to one another. By having similar words in their text, we will get similar vectors and therefore our clustering algorithm will cluster them together.

2.4 Dimensionality Reduction

After the TF-IDF transformation, the data is now in a very high dimensional space which causes instability when doing clustering. When working with large amounts of dimensions many problems can occur with finding appropriate distances between each other. This becomes a problem when using clustering algorithms such as K-Means.

To bring our data down to a lower number of dimensions I decided to use Truncated Singular Value Decomposition or (SVD). Also known as Latent Semantic Analysis in text mining (scikit-learn, n.d.). I used L2 norm on the vectors before using this. L2 norm will ensure that your vectors only compare directions and not their size. If we didn't do this then longer abstracts would have more of an effect when calculating distance between documents.

I played around with many different values and found that the lower number of dimensions gave me a better result. I tried everything from 50 to 300 components. If I could get a good explained variance I wanted to find what gave me a more descriptive cluster.

So, I decided on 50 components to be my final decision. After seeing my results and using some of my previous measures such as the Silhouette score, I felt that this was the best amount. Although we have a low explained variance (around 8%) this is normal for high dimensional data sets that deal with text. Scikit learns documentation shows us that when using LSA for text clustering we should expect a low amount of variance. Their example had 18.4% with 100 components (scikit-learn, n.d.).

For the 2D visualizations (at the end of the notebook), I reduced the data one more step from 50 dimensions down to 2 using PCA. Allowing me to visualize the clusters and gain some understanding. Knowing that I was going to lose information from these vectors.

I also tried to use Multi-Dimensional Scaling (MDS) for my visual. Which allows me to project points that have similar distances to one another. I figured this may have been better suited for my problem, but I did not get any good results from my projection.

2.5 Clustering Model

For clustering, I mainly compared two different approaches: K-Means and Gaussian Mixture Models (GMM).

However, Hierarchical Clustering was something I considered, mainly because it is intuitive and gives tree-like structure. But I had in mind that the computational cost grows as $O(n^2)$ and with 150k documents, this would have made the clustering process extremely slow and computationally expensive.

Then I moved on to K-Means and GMM. As for K-Means I wanted to find out what number of clusters would be optimal. Using the traditional ways of elbow and silhouette score. Elbow method wasn't too helpful as you can see the inertia just decreases little by little (Figure 2). Silhouette score helped a bit more (Figure 3). It showed that k should be 3 which wasn't helpful when we are trying to find clusters of certain fields to work with. With only 3 clusters I would basically end up with very broad groups like 'Computer Science', 'Mathematics' and 'Physics'.

So instead, I started to iterate manually and explored higher values of k: 8, 10 and 12. The decision I was trying to make was based on metrics and on how interpretable the clusters were for the company. I would run the model, look at the top words in each cluster and check whether they formed meaningful scientific topics or if everything was getting mixed.

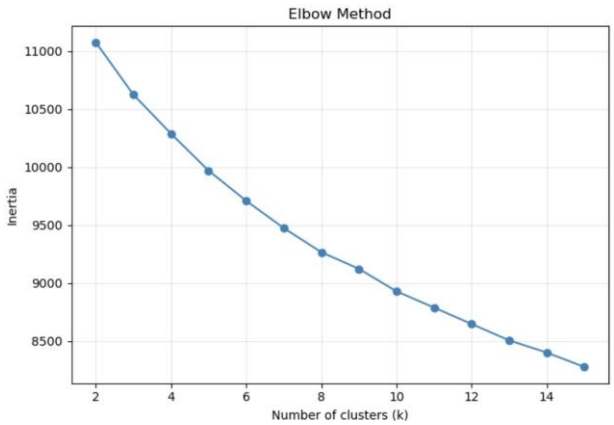


Figure 2: Elbow Method for k=2 to 15

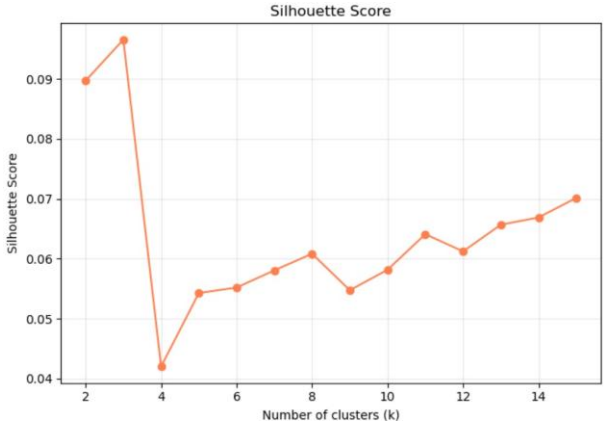


Figure 3: Silhouette Score for k=2 to 15

When using k=8 we can start seeing issues where we want certain topics to be separated but they're all combined into one cluster. Some examples would be Deep Learning and Systems and Control. Computer Vision isn't really separated from Machine Learning either. There's a loss of information that could be useful.

When increasing to k=10 we can see some nice areas such as Optimization or vision. But there are some key topics that we want to see separated and aren't. Black Holes and Galaxies are still joined. And we can't separate Large Languages Models (LLMs) from other Generative AI models.

Finally, k=12 seems to be the best choice. It gives us a slightly higher silhouette score and allows us to see many different areas we may be interested in. Such as LLMs, Generative AI, Optimization, Quantum Computing or Black Holes. Which would be great for collaboration.

I tried GMM with k=12 as well. I figured since it allowed the clusters to have more freedom it may give better results. But this was not the case. Silhouette score was very low (~0.0063) and clusters were very dirty. It was difficult to make sense of the results.

GMM tends to not work very well when working in higher dimensions. As your number of dimensions grow your covariance matrix will grow. Causing under fitting and bad clustering (Zhao et al., 2018). Another reason GMM may not work with our data set is because it assumes the data points are gaussian distributed.

Therefore, I will use K-Means as my final model with k=12.

Table 1 summarizes the final configuration selected after a parameter search:

Parameter	Value
Date range	2023-2025
Sample size	150,000 papers
TF-IDF max_features	8,000
TF-IDF min_df	10
TF-IDF max_df	0.8
SVD components	50
Final k	12
Silhouette (k=12)	0.0639

Table 1: Final Configuration

3. What the Data Reveal About Current Science

3.1 The 12 Research Clusters

After finishing the last model: Do the clusters have any meaning? Are they helpful to the business?

To give a name to each cluster I mainly analysed the top 10 words that represented each one. Also looking at what was the most common arxiv category in each cluster. I wanted to have names that described the topic and would make sense if someone was collaborating on those papers.

Below I will show you a table that contains our last 12 clusters. With the name of the topic, the most representative words and the number of papers they contain. As well as their percentage of the total sample and what arxiv category they fall under. Each cluster has a different size this shows how science can be uneven in certain fields.

As we can see our largest cluster is Systems & Control it makes up close to 21% of our sample. This cluster is the most general one that we got from our data. It can represent many applied engineering or applied machine learning papers. So, it's a bit broad as a cluster.

Next, we have our other 2 large clusters which are more classical fields of study. Condensed matter physics and Mathematics. These fields have been around for a long time, so it makes sense that our clusters are bigger. It shows that our model is picking up on older fields of study.

I think it's cool to see that the AI clusters are smaller but more precise. LLMs and generative AI are not taking up much space in our clusters. But when you look at the keywords they are all about these topics. So, they could be our fastest growing cluster. On the opposite end we have black holes and relativity which is our smallest cluster. We have almost all the papers under gr-qc. Which shows that our model can detect certain fields of study.

Cluster	Topic	Key words	Papers	Share	Main arXiv category
1	Quantum computing	quantum, states, entanglement, qubits, circuit	5,745	3.83%	quant-ph
2	Astrophysics (stars & galaxies)	galaxies, stars, mass, stellar, star	6,639	4.43%	astro-ph.GA
3	Mathematics (algebra and geometry)	group, space, theory, finite, groups	21,689	14.46%	math.NT
4	Systems & control	systems, model, control, framework, design	31,467	20.98%	cs.RO
5	Large Language Models (LLMs)	llms, language, models, reasoning, tasks	8,929	5.95%	cs.CL
6	Graph theory	graph, graphs, vertices, edges, node	3,338	2.23%	math.CO
7	Black holes & relativity	black, hole, gravitational, gravity, horizon	1,497	1.00%	gr-qc
8	Deep learning	data, learning, model, neural, training	17,201	11.47%	cs.LG
9	Condensed matter physics	magnetic, phase, energy, spin, temperature	25,759	17.17%	hep-ph
10	Generative AI & intelligent agents	ai, intelligence, artificial, generative, agents	1,849	1.23%	cs.CY
11	Computer vision	image, images, segmentation, video, visual	12,772	8.51%	cs.CV
12	Optimization & numerical methods	problem, solutions, convergence, optimization, numerical	13,114	8.74%	math.OC

Table 2: The 12 Clusters

Overall, the results provide an overview of the current distribution of scientific research areas. With some very large and stable fields, some broad applied areas and a few smaller but very fast AI topics.

3.2 Topic Space Visualization

To get a better idea of what our data looks like we can reduce the data to 2 dimensions using PCA which was shown in point 2.4. Of course, this loses us a lot of information but I wanted to be able to visualize the data. When looking at the data it shows a L - shaped pattern.

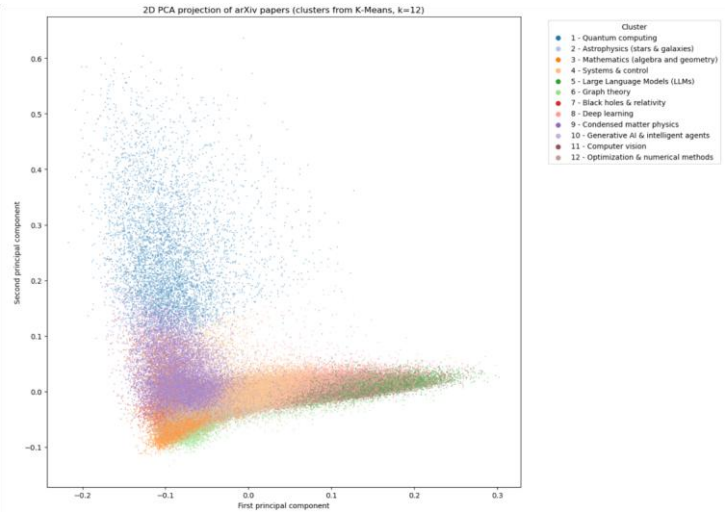


Figure 4: Scatter Plot – 2D Topic Space Visualization

On the top of the plot, we have Quantum Computing and Astrophysics these are pretty far away from everything else. Which I believe can be attributed to them having more unique words.

Somewhere in the middle we see Condensed Matter Physics, Black Holes and Math all closer together as they all deal with similar concepts and words.

The left-hand side seems to be where all the computational fields fall under. We have LLMs, Computer Vision, Optimization, Systems & Control, GenAI etc. Its more crowded and jumbled together which makes sense for computational topics. As they all share many words.

I think this is interesting from a business perspective as well. As we seem to have 2 worlds here one more physical and theory based. And one more computation/AI focused.

Although its just a 2D map it gives us a little bit of insight that maybe the clusters are true.

3.3 Temporal Evolution of Research Topics

I looked at how the clusters evolve over time between 2023 and 2025. This part is more than relevant since I get to understand where things are heading.

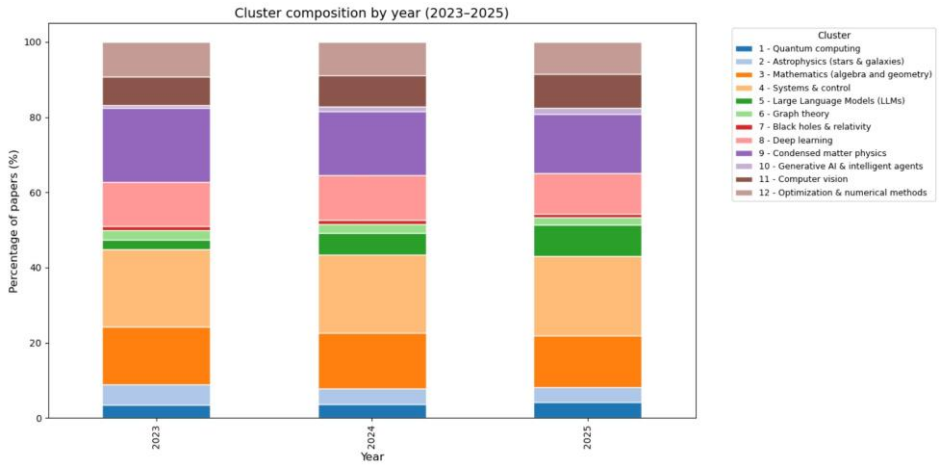


Figure 5: Temporal Evolution Bar Chart of Cluster Sizes

Probably the most obvious one is that AI topics are on the rise. LLMs increased the most from 2.53% in 2023 to 8.17% in 2025. Which is a significant amount in such a short span of time. Generative AI is also increasing from 0.69% to 1.55%. Although not much it's going up. Computer vision increased as well just by a smaller margin. Going from 7.66% to 9.15%. It isn't growing as much as LLM's but still has been trending upwards.

We can see Systems & Control and Quantum computing increasing slightly as well over the given time frame. So, they are doing well too.

There are some topics that we can see declining over time. Condensed matter physics went from 19.69% to 15.74%. Though still the highest cluster of topics. Math went from 15.28% to 13.75%, and astrophysics went from 5.41% to 3.90%. This could be because of all the AI topics growing. But other clusters remained relatively the same. Graph theory and black holes and relativity had small fluctuations over time.

4. Conclusions and Recommendations

What we can infer from these results is that there are at least 2 categories of opportunities. One is the big well-known areas of science such as mathematics physics and parts of engineering. These will always be around and can provide great long-term collaborations if the company wishes to collaborate on a higher level of academics. The other category would be all areas surrounding AI. With LLMs Generative AI and Computer vision. These fields are smaller at the moment, but they are increasing at a faster rate. If the company wants to push towards newer directions, these are what they should look at.

We were able to find 12 topics that have similar characteristics. All of them seem to be good areas of study.

As stated, before all areas related to AI are increasing at a greater rate then the rest. Areas such as LLMs Generative AI and Computer vision have increased over the entire period. LLMs being the largest by relative share. Which is quite significant.

Some topics that have been studied for a longer time are starting to decrease in share. I don't think they are decreasing but other topics are surpassing them. Such as AI.

One thing we can also note is that when looking at our 2D plot the physical science and computation/AI areas were spread out in different areas. Which further proves that our clusters are accurate.

There are some topics that the company should focus on. These are all high priority. Topics that have to do with AI are what they should investigate. Such as LLMs Generative AI and Computer Vision. These topics have all shown significant growth and are close to many fields of study. Quantum computing can also be considered. Though its not as big as the others it is unique. Which can put the company at a higher tier of research.

Other topics the company can look into would be condensed matter physics, mathematics and deep learning. These topics can also provide great collaborations. But they aren't as pressing as the ones mentioned above. If the company doesn't want to go for more AI driven topics, they can investigate these. Overall, the company should look into AI topics.

5. Limitations and Improvements

Limitations to the analysis:

I only analysed abstracts not full texts. This allows us to pick up on general themes, but we lose out on more specific information. The second limitation is that when using SVD we can reduce to only 8% of explained variance. Although this is not too uncommon when analysing text data. But it does mean that some information is being reduced.

Another thing is that I only chose 150,000 papers at random. There could be some slight difference if I had sampled another 150,000 papers but the overall trend looks quite clear so I feel as though the analysis would be similar.

Future Analysis:

There are many things that could be done to further analyse the data. Embeddings could be used to see how well they can pick up on information compared to TFIDF. Or even analysing full text papers.

Overall, I think this is a good start to seeing what science is currently focusing on and there could be more detailed information found if we did another round of analysis.

Code can be found on my GitHub here:

https://github.com/Devgita/Categorizing_Trends_in_Science

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